Qinyan Liu

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Education

University of Science and Technology of China (USTC), School of the Gifted Young Sep 2023 – Jul 2027

B.S. in Mathematics (Expected), Specialization: Probability Theory and Mathematical Statistics

GPA: 4.08/4.30 (93.1/100) | **Rank: 2/64** in specialization, 5/137 in Mathematics

Publications

Qinyan Liu, Yan Chen, Canhong Wen.

"Transformers as Provable Approximators of Sparse Principal Component Analysis." [OpenReview] [Code] [Poster]

Proceedings of STAI-X 2026: Statistics and Trustworthy AI for Cross (X)-Domain Acceleration, 2026.

Manuscripts Under Review

Qinyan Liu, Shihao Wu, Gongjun Xu, Ji Zhu.

"Generative Modeling of Hypergraphs Using Latent Hyperbolic Geometry."

Under review at ICLR 2027.

Research Experience

Generative Modeling of Hypergraphs in Hyperbolic Space May 2026 – Present

Advisors: Prof. Gongjun Xu and Prof. Ji Zhu, Department of Statistics, University of Michigan

- Developed a hyperbolic latent-space generative model for hypergraphs and a scalable two-stage procedure combining likelihood-based embedding with empirical bootstrap or Riemannian diffusion.
- Contributed substantially to the theoretical analysis of latent geometry identifiability and generative consistency.
- Evaluated the method through simulation studies and experiments on three real-world hypergraph datasets, including model tuning, extensive baseline comparisons, and privacy-related diagnostics.
- Achieved leading generation fidelity with a parsimonious two-dimensional hyperbolic representation and demonstrated coherent latent structure and domain-plausible generated hyperlinks.

Transformers for Sparse Principal Component Analysis Jun 2025 – Present

Advisor: Prof. Canhong Wen, Department of Statistics and Finance, USTC

- Proved that multi-head Transformers can implement Sparse Principal Component Analysis (SPCA) under noise, extending existing theoretical work on Transformer approximation to the fully unsupervised setting.
- Designed parameter constructions aligned with iterative Truncated Power updates, and derived guarantees by decomposing recovery error into algorithmic approximation, statistical estimation, and generalization components.
- Conducted experiments on synthetic and high-dimensional gene-expression data, achieving competitive performance against classical baselines, with extensive ablations across model configurations and noise levels.
- Demonstrated efficient batched inference after training, highlighting amortized Transformer-based SPCA inference.
- Extending this framework toward data-adaptive selection of the number of components and sparsity levels.

Leveraging Zero-Inflated Data via Double Machine Learning Jul 2025 – Oct 2025

Advisor: Prof. Kan Xu, W. P. Carey School of Business, Arizona State University

- Studied treatment effect estimation in real-world online marketing data with zero-inflated and censored demand.
- Implemented preliminary Double Machine Learning and doubly robust estimators to study sparse click outcomes.
- Investigated zero-inflated models and semiparametric methods for treatment effect estimation.

Selected Coursework

Probability & Statistics

Probability Theory (97)	Regression Analysis (96)	Applied Stochastic Processes (92)
Probability Theory-Outer Chapter (91)	Mathematical Statistics (90)	Multivariate Analysis (92)

Mathematics

Mathematical Analysis I – III (90 – 92)	Linear Algebra I – II (92 – 95)	Differential Geometry (95)
Real Analysis (Honor) (88)	Complex Analysis (90)	Differential Equations (92)

Optimization, programming and ML

Computer Programming A (98)	Operations Research (90)	Foundations of AI and ML (ongoing)

Honors & Awards

- **Guomo Ruo Scholarship** Candidate (highest undergraduate honor at USTC) Nov 2026
- **China National Scholarship** (highest scholarship from Ministry of Education of China) Dec 2025
- **China National Scholarship** Dec 2024
- Outstanding Freshman Scholarship, Gold (**top 3%**) Jan 2024
- National Undergraduate Mathematics Competition, First Prize (Provincial) Dec 2024
- FLTRP • ETIC Cup National English Reading Competition, Gold Award (**1st Place Nationwide**) Dec 2023

Teaching

Teaching Assistant, Linear Algebra II, USTC; Instructor: Prof. Xinmao Wang, School of Mathematical Sciences

- Graded assignments and answered students' questions.
- Led problem-solving sessions and delivered review lectures.

Academic Talks

SLDS 2026 (Poster Presentation), Conference on Statistical Learning and Data Science, New York, NY; "Transformers as Provable Approximators of Sparse Principal Component Analysis"

Loo-Keng Hua Seminar (H), USTC; "Provable Transformers for Sparse PCA"; final grade: **100/100**

Skills

Programming: Python (PyTorch, JAX, NumPy, SciPy, pandas, scikit-learn, Gurobi), R, C, Mathematica

Systems & Research Tools: Linux, Bash, Slurm, CUDA, SSH, Git, LaTeX

Languages: Chinese (native); English (fluent)

Standardized Tests: TOEFL **113**/120 (R29, L30, **S27**, W27); GRE **333**/340 (Q170, V163)

Leadership & Activities

Vice President & WeChat Public Account Manager, USTC Student Chorus

- Led event organization and coordinated publicity efforts for the chorus's 20th anniversary.
- Received the USTC Outstanding Student Organization Leader honor (Jun 2026).

Member of the Student Union, School of the Gifted Young, USTC

- Organized cultural, athletic, and community events to enrich students' campus life.
- Received the USTC Outstanding Student Leader honor (Apr 2025).